

RESIDUAL STRENGTH AFTER TRANSVERSE BVID IMPACT

Section: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3405**

Title: **RESIDUAL STRENGTH AFTER
TRANSVERSE BVID IMPACT**

Revision: **1.1**

Date: **11/12/2024**

1 INTRODUCTION

1.1 OBJECT

This chapter presents the applicable failure criterion for obtaining the static strength margin of composite parts including undetected impact damages. The failure criterion for the residual strength after (low velocity) BVID impacts in composite materials is defined in the following reference:

ME-T-NT-130045 Is. A "FAILURE CRITERION: RESIDUAL STRENGTH AFTER BVID IMPACT IN COMPOSITE PARTS (DAMAGE TOLERANCE)"

1.2 SCOPE AND LIMITATIONS

The following paragraphs are just an extract of above mentioned document, applicable to solid FRP laminates.

Treatment of barely visible damages caused by edge impacts are still not covered by this document.

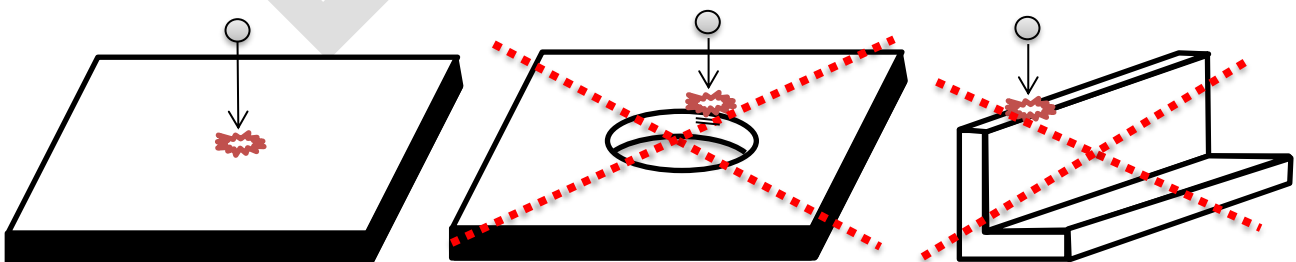


Figure 1. Transversal impact and edge impacts (for later case method TBD)

RESIDUAL STRENGTH AFTER TRANSVERSE BVID IMPACT

2 METHOD DESCRIPTION

2.1 DAMAGE TOLERANCE IN SOLID LAMINATES

Failure criterion

Maximum strain criterion will be used to obtain the static margin, comparing allowable design values derived from test coupons (TAI/CAI) and critical strain level in fibre direction:

$$RF = TAI / \varepsilon \quad \dots \text{if } \varepsilon > 0$$

$$RF = | CAI / \varepsilon | \quad \dots \text{if } \varepsilon < 0$$

Interaction with plate bending

In case bending loads in laminate are significant, the margin calculated according to method above can be increased by taking advantage of BAI design values. The typical procedure in such cases is to obtain strain values corresponding to membrane (in-plane) loads on one side (ε_{mem}), and strain values corresponding to bending (out-plane) loads on the other side (ε_{ben}). Notice that the strain in weft direction of UD tape plies (resin direction) shall NOT be used. Then, the RF is obtained by a linear simple interaction of membrane and bending ratios, taking advantage of specific Bending After Impact (BAI) allowable values.

$$RF = 1 / [| \varepsilon_{mem} / (TAI, CAI) | + | \varepsilon_{ben} / BAI |]$$

Design allowable values

Design allowable values TAI, CAI (and BAI) will be defined for each project according its particular design and certification requirements (materials and processes, BVID threshold definition, environmental conditions, ...). Stress Methods department shall define, or support the definition, of these values in order to use harmonized and consistent values within Airbus DS projects.

Note that, while TAI strength shall be typically set as a scalar value for all laminates made of one material (thus a material allowable), current models for CAI are defined with sensitivity to laminate thickness and directionality (see chapter §1421). One of the implications of this is that the CAI allowable to get the reserve factor for each ply/direction shall be specific of the direction analysed, i.e. one value at X / 0°, another for Y / 90° and another for ±45° directions, in laminates following the classical design rules built on the legacy/quad angles base (0°, 45°, 90°, -45°).

In the absence of more precise test data it is recommended to use TAI as cut-off value for BAI.

Strength check process

The recommended implementation of this method, provided it has to be applied onto a laminate following the classical design rules (including mid-plane symmetry), its stiffness matrices calculated according to the *Classical Laminates Theory*, known its residual strengths, i.e. CAI₀, CAI₉₀, CAI₄₅, TAI,

and BAI (typ. equal to TAI), which is subjected to a set of plate forces (N_x , N_y , N_{xy} , M_x , M_y , M_{xy}) shall be to proceed as follows:

- Calculate the mid-plane strain (ϵ_{xx} , ϵ_{yy} , γ_{xy}) of the laminate when subjected to (only) in-plane loads (N_x , N_y , N_{xy}). See chapter §3401 for advice on the *Classical Laminates Theory*.

- Calculate also the mid-plane strains at 45° and -45° directions, eg. using the expressions:

$$\epsilon_{45} = (\epsilon_{xx} + \epsilon_{yy} + \gamma_{xy}) / 2 \quad \epsilon_{-45} = (\epsilon_{xx} + \epsilon_{yy} - \gamma_{xy}) / 2$$

... then membrane strains in the 4 directions of interest shall be:

$$(\epsilon_0 (= \epsilon_{xx}), \epsilon_{90} (= \epsilon_{yy}), \epsilon_{45}, \epsilon_{-45})_{mem}$$

- Calculate the mid-plane curvature (κ_{xx} , κ_{yy} , κ_{xy}) of the laminate when subjected to (only) moments (M_x , M_y , M_{xy}), and the associated strains at laminate faces (at distance $\pm t / 2$ from laminate mid-plane):

$$\epsilon_{xx} = \pm \kappa_{xx} \cdot t / 2, \quad \epsilon_{yy} = \pm \kappa_{yy} \cdot t / 2, \quad \gamma_{xy} = \pm \kappa_{xy} \cdot t / 2$$

... and the bending strains in the 4 directions of interest (use the abovementioned expressions for ϵ_{45} , ϵ_{-45}) :

$$(\epsilon_0 (= \epsilon_{xx}), \epsilon_{90} (= \epsilon_{yy}), \epsilon_{45}, \epsilon_{-45})_{ben}$$

- Then get the RF as the minimum of those calculated for the 4 directions:

$$RF_0 = 1 / [|\epsilon_{0mem} / (TAI, CAI_0)| + |\epsilon_{0ben} / BAI|]$$

$$RF_{90} = 1 / [|\epsilon_{90mem} / (TAI, CAI_{90})| + |\epsilon_{90ben} / BAI|]$$

$$RF_{45} = 1 / [|\epsilon_{45mem} / (TAI, CAI_{45})| + |\epsilon_{45ben} / BAI|]$$

$$RF_{-45} = 1 / [|\epsilon_{-45mem} / (TAI, CAI_{-45})| + |\epsilon_{-45ben} / BAI|]$$

$$RF = \min (RF_0, RF_{90}, RF_{45}, RF_{-45})$$

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
ϵ , ϵ_{mem}	-	In-plane (extensional) strain
ϵ_{ben}	-	Bending strain
BAI	-	Bending After Impact. Strength of laminate in plate bending including an undetected damage in the threshold of visibility.
$BVID$	-	Barely Visual Impact Damage. Impact damage in the threshold of detectability.
CAI	-	Compression After Impact. Compression strength of laminate including an undetected damage in the threshold of visibility.
DT	-	Damage Tolerance (in this scope refers to the residual strength after BVID impact)
FRP	-	Fibre Reinforced Polymer
M_x	Nmm/mm	Bending X moment in plate (per unit length)

M_y	Nmm/mm	Bending Y moment in plate (per unit length)
M_{xy}	Nmm/mm	Twisting moment in plate (per unit length)
N_x	N/mm	Longitudinal X load in plate (per unit length)
N_y	N/mm	Transverse Y load in plate (per unit length)
N_{xy}	N/mm	In-plane shear load in plate (per unit length)

RF - Reserve Factor

TAI - Tension After Impact. Tension strength of laminate including an undetected damage in the threshold of visibility.

REFERENCES

	Document Reference	Issue	Title
Ref. 1	ME-T-NT-130045	Is.	A FAILURE CRITERION: RESIDUAL STRENGTH AFTER BVID IMPACT IN COMPOSITE PARTS (DAMAGE TOLERANCE)

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
1.0 30/04/2020	G. Baños	First issue	All pages
1.1 11/12/2024	G. Baños	Detailed description of analysis strength check process detailed	Page 2-3

Table 1. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods & Optimization	G. Baños 11/12/2024	-
Structural Analysis	-	F. Glock XX/12/2024

Table 2. Approval signatures table for last revision of this chapter